

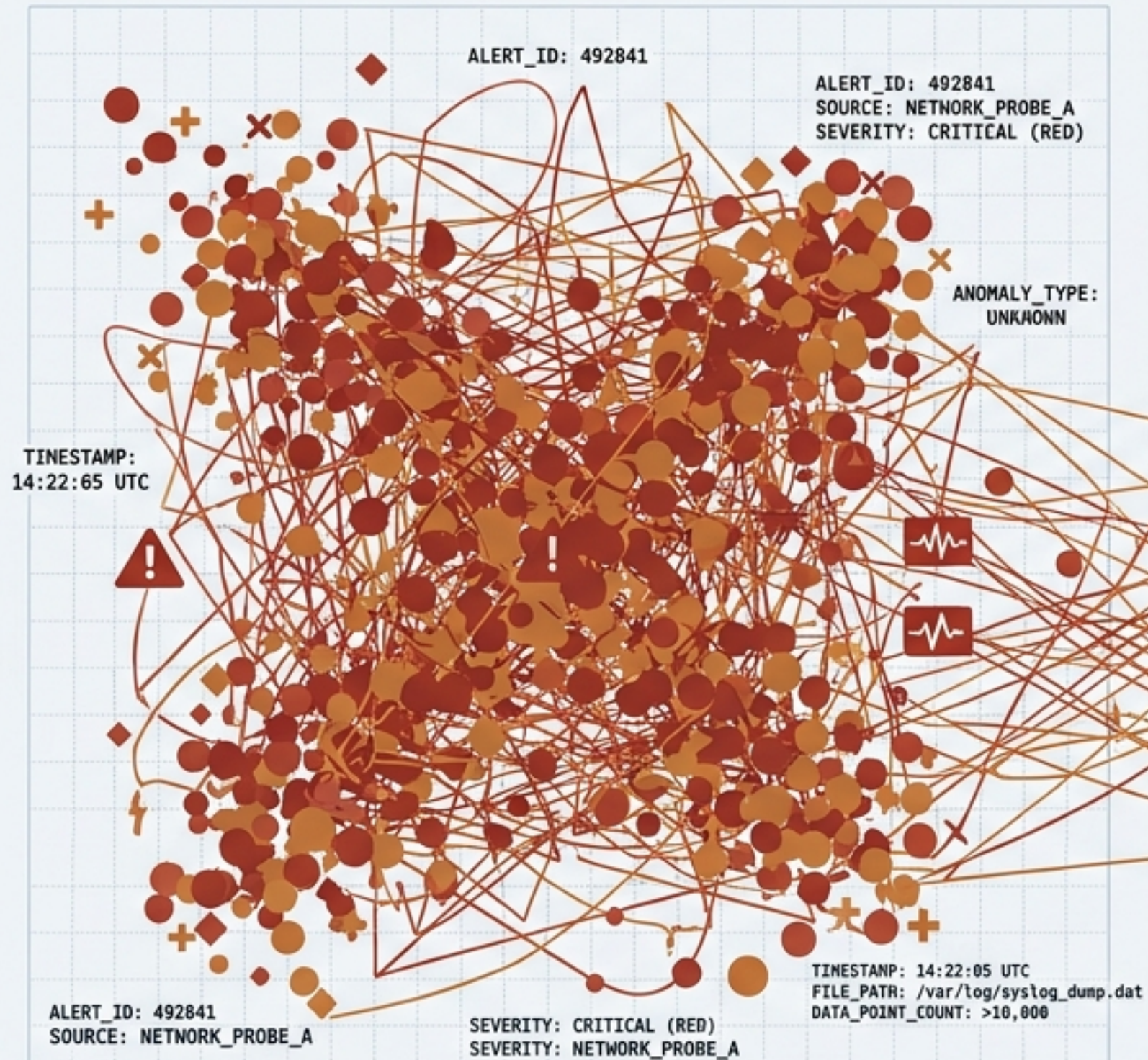
ThreatSense AI

AI-Powered Threat Intelligence Correlation & Alert Prioritisation

Hackathon:	IBM BoB AI Innovation Hackathon 2026
Problem:	D2 – Threat Intelligence Correlation Assistant
Track:	AI
Team:	CyberGuard

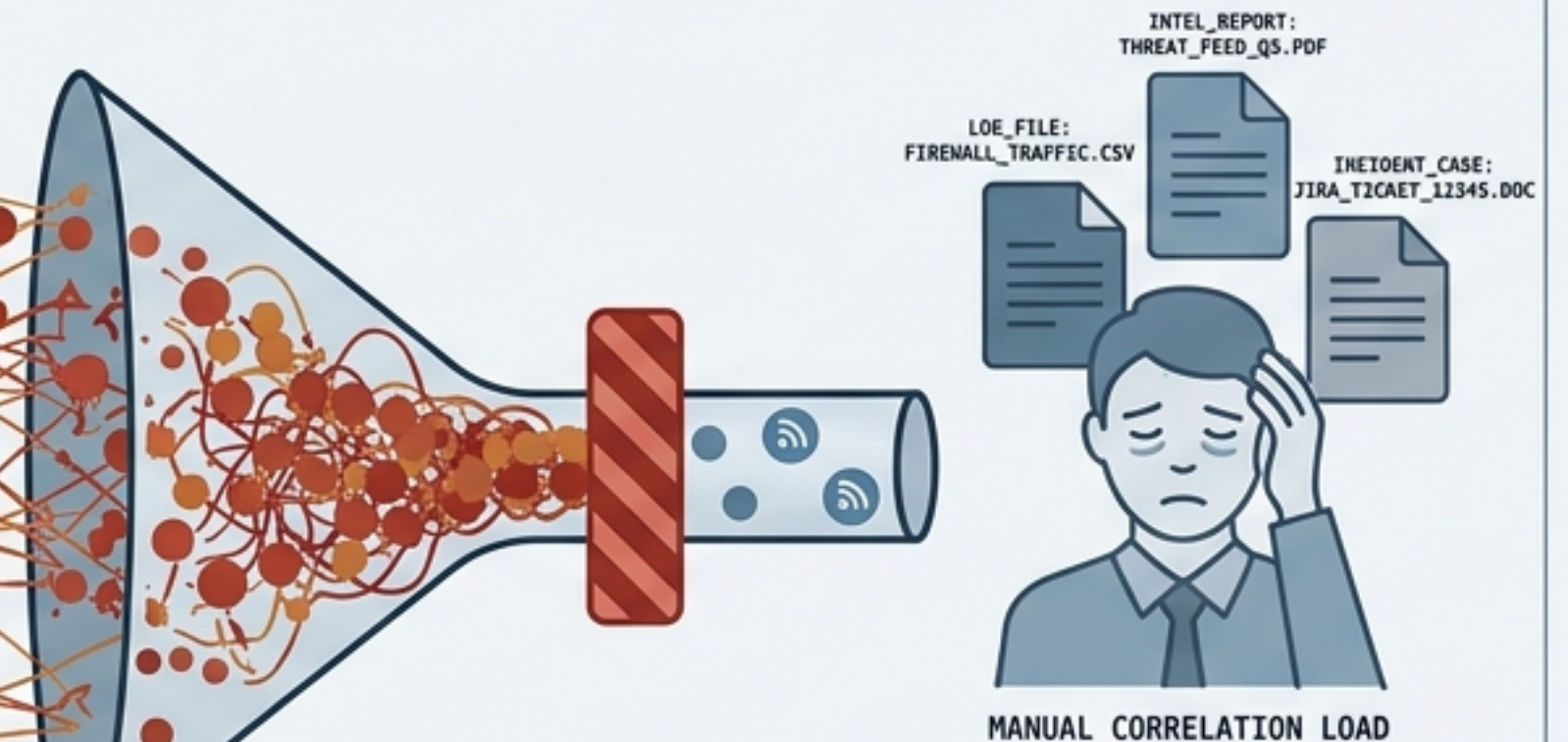
The Analyst's Dilemma: Signal Lost in the Noise

OVERWHELMING ALERT VOLUME & NOISE

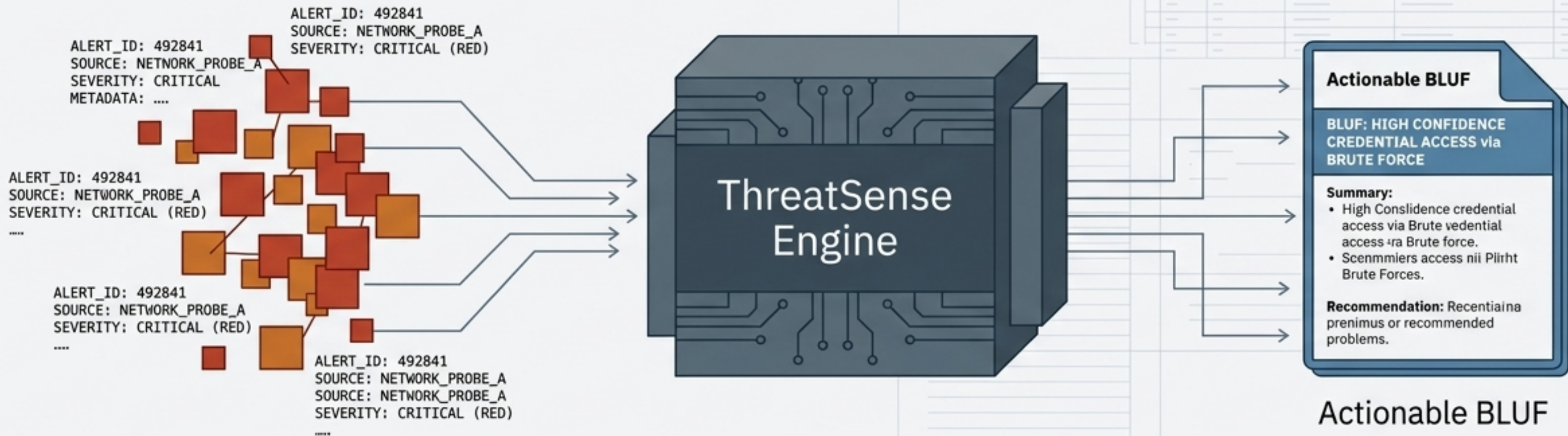


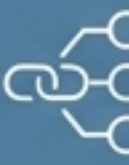
The bottleneck is not data collection; it is manual correlation.

- Analysts are overwhelmed by high-volume, heterogeneous alerts.
- Manual pivoting is required to distinguish genuine threats from false positives.
- Preparing investigation summaries manually delays critical incident response.



Automating the Triage Pipeline

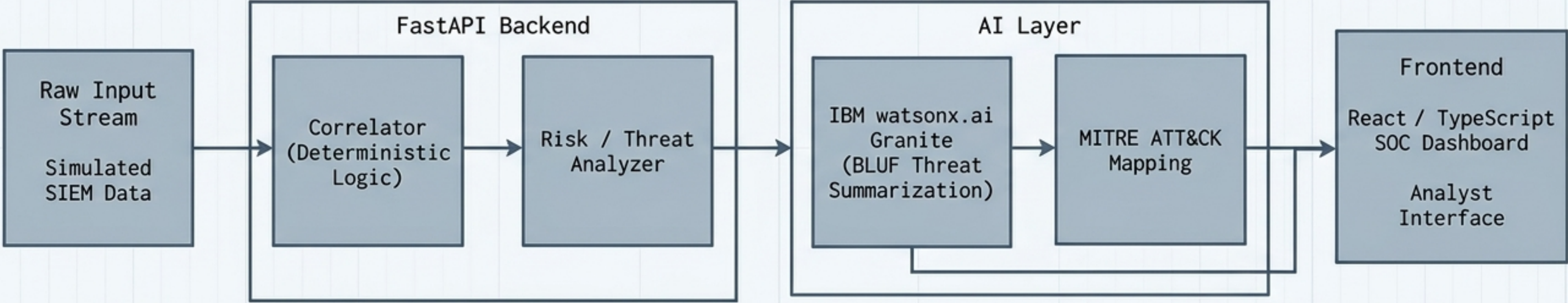


			
Unify Ingests and normalizes multi-source feeds (SIEM, EDR, Email).	Correlate Mathematically groups alerts by IP, attack pattern, and time.	Contextualize Maps behavior to MITRE ATT&CK techniques automatically.	Summarize Generates natural-language Bottom Line Up Front (BLUF) summaries using IBM watsonx.ai Granite.

Workflow Transformation Matrix

	Traditional SOC	✔ ThreatSense AI ✔
Correlation	Manual IP matching across multiple browser tabs.	Automated time-window and IP heuristic correlation.
Prioritization	Static, rule-based severity tags.	Dynamic 0-100 risk scoring based on multiple weighted variables.
Threat Context	Scattered intelligence requiring external lookups.	Native MITRE ATT&CK technique mapping.
Reporting	Tedious manual incident summary writing.	Generative AI BLUF summaries via IBM Granite.

System Architecture & Data Pipeline



Spitzmeldung	Date				Time
16.09	2024-05-08				Dem
16.26	2024-05-07				Dem
16.09	2024-05-06				Dem
16.15	2024-05-07				Dem
16.15	2024-05-07				Dem
16.15	2024-05-07				Dem
16.16	2024-05-07				Dem
16.17	2024-05-07				Dem
16.18	2024-05-07				Dem
16.19	2024-05-07				Dem
16.20	2024-05-07				Dem
16.21	2024-05-07				Dem
16.22	2024-05-07				Dem

Detected Source	21	0	20
Detected Source	21	2	20
Detected Source	15	0	10
Detected Source	22	6	10
Detected Source	23	5	20
Detected Source	25	0	10
Detected Source	27	0	10
Detected Source	29	10	20
Detected Source	28	0	21
Detected Source	20	10	20
Detected Source	20	0	20
Detected Source	20	10	22

1000	1000-07-02	0	10-10-11
1001	1001-07-03	11	10-10-11
1002	1002-07-03	0	10-10-11
1003	1003-07-03	0	10-10-11
1004	1004-07-03	0	10-10-11
1005	1005-07-03	11	10-10-11
1006	1006-07-03	0	10-10-11
1007	1007-07-03	0	10-10-11

Taming the Alert Queue

Immediate Global Triage

Real-time visibility into Total (50), High (30), Medium (15), and Low (5) severity alerts.

TOTAL ALERTS		HIGH SEVERITY		MEDIUM SEVERITY		LOW SEVERITY	
50		30		15		5	
Search alerts...				All Severities		All Sources	
ID	SOURCE	TINEESTAMP	SRC IP		DST IP		
ALT-1814	NetFlow-Analyser	14/09/2026, 20:56:43	18.6.2.28		18.6.2.20		
ALT-1841	SIEM-QRadar	14/09/2026, 20:46:43	153.45.12.88		18.6.10.20		
ALT-1843	Email-Gateway	14/09/2026, 20:46:43	18.6.2.25		18.6.1.10		
ALT-1815	SIEM-QRadar	14/09/2026, 20:45:43	18.6.2.25		18.6.2.20		
ALT-1816	Firewall-IDS	14/09/2026, 20:42:42	18.6.2.28		18.6.2.20		
ALT-1865	EDR-Sentinel	14/09/2026, 20:34:48	10.8.18.20		18.6.10.15		
ALT-1864	WAF-CloudEdge	14/09/2026, 20:25:45	155.228.151.42		18.6.10.10		
ALT-1852	Firewall-IDS	14/09/2026, 20:25:45	155.228.151.42		18.6.10.20		
ALT-1889	Firewall-IDS	14/09/2026, 19:56:43	10.6.18.5		91.188.36.77		

TS

ThreatSense AI

Threat Intelligence & Alert Prioritization

50

TOTAL ALERTS

30

HIGH SEVERITY

15

MEDIUM SEVERITY

5

LOW SEVERITY

38

OPEN

4

IN REVIEW

8

CLOSED

Search Alerts...

All Severities

All Sources

All Statuses

ID	SOURCE	TIMESTAMP	SRC IP	DST IP	ALERT TYPE	SEV	STATUS
ALT-1814	NetFlow-Analyser	14/09/2026, 20:56:49	18.6.2.28	18.6.2.20	Ransomware Behavior De...	MED	Open
ALT-1841	SIEM-QRadar	14/09/2026, 20:46:43	153.45.12.88	18.6.10.20	Anomalous Login Location	MEDIUM	Closed
ALT-1843	Email-Gateway	14/09/2026, 20:46:43	18.6.2.25	18.6.1.10	Data Exfiltration Suspected	MED	Open
ALT-1815	SIEM-QRadar	14/09/2026, 20:45:43	18.6.2.28	18.6.2.20	Malware Signature Match	MEDIUM	Open
ALT-1816	Firewall-IDS	14/09/2026, 20:42:42	18.6.2.28	18.6.2.20	Suspicious PowerShell Exe...	MEDIUM	Open
ALT-1865	EDR-Sentinel	14/09/2026, 20:34:48	10.8.18.20	18.6.10.15	Privilege Escalation	MED	Open
ALT-1864	WAF-CloudEdge	14/09/2026, 20:25:45	155.228.151.42	18.6.10.10	Lateral Movement Detected	MED	Open
ALT-1852	Email-Gateway	14/09/2026, 20:25:45	155.228.151.42	18.6.10.20	Brute Force Login Attempt	MED	Open
ALT-1889	Firewall-IDS	14/09/2026, 19:56:43	10.6.18.5	91.188.36.77	Data Exfiltration Suspected	MED	Open

Select an alert to see detailed analysis

Showing 10 of 50 alerts

Malware Signature Match

MEDIUM

Open

Granular Investigation

Filter by Severity, Source, or Status to cut through noise.

Unified Schema

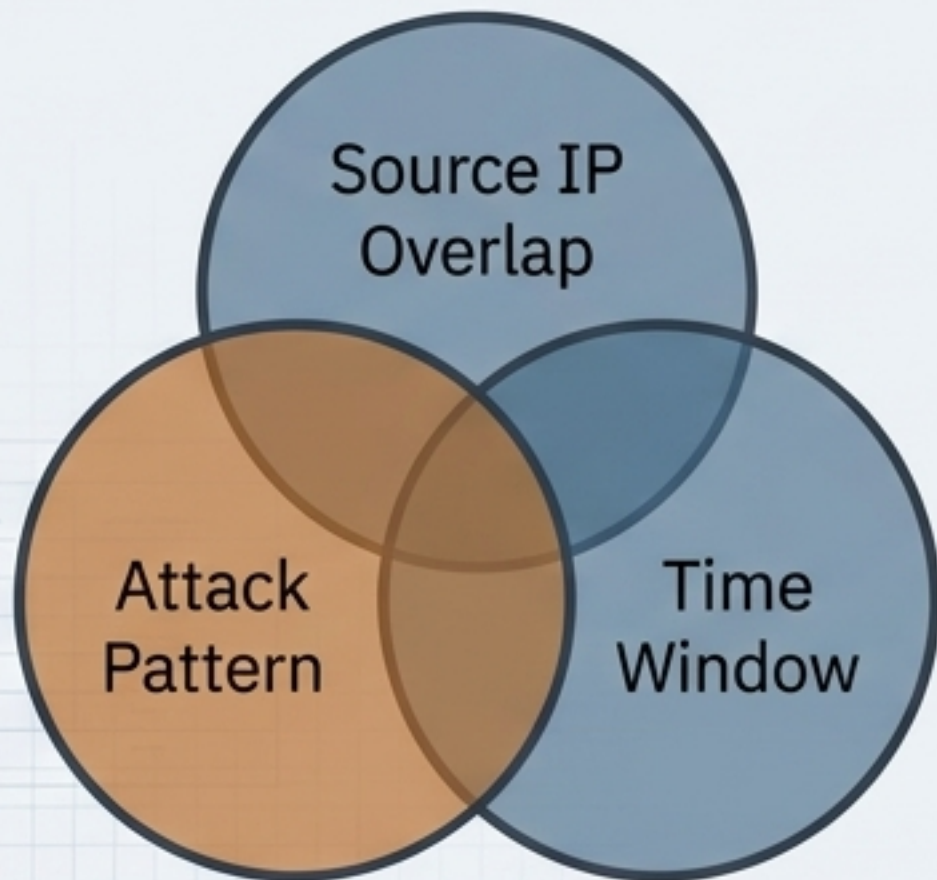
Disparate sources (NetFlow, SIEM-QRadar, Email-Gateway, EDR-Sentinel) normalized into a single readable event log.

ID	SOURCE	TIMESTAMP	SRC IP	ALERT TYPE	SEV	STATUS
ALT-1814	NetFlow-Analyser	14/09/2026, 20:56:49	18.6.2.28	Ransomware Behaviour Be...	MED	Open
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ALT-1852	Firewall-IDS	14/09/2026, 20:25:45	155.228.151.42	Brute Force Login Attempt	MED	Open
ALT-1889	Firewall-IDS	14/09/2026, 19:56:43	10.6.18.5	Data Exfiltration Suspected	MED	Open

The Intelligence Engine: Correlation & Scoring

Correlation Heuristics

Alerts are currently grouped using deterministic time-window and IP-overlap logic to ensure predictable, transparent clustering.



Risk Score Anatomy (0–100)

Base Severity

+ Event Frequency

+ Asset Criticality

+ MITRE Coverage

= **0–100 Risk Score**

Actionable Intelligence: Context & Mitigation

Automated Framework Mapping

Instantly identifies behaviors like T1486 (Data Encrypted for Impact) and T1059.001 (PowerShell) to provide attacker context without manual pivoting.

Accelerated Incident Response

Generates clear, immediate mitigation steps (e.g., Isolate host, Do NOT reboot) to preserve forensics and halt lateral movement.

MITRE ATT&CK TECHNIQUES

- T1486 Data Encrypted for Impact (Impact)
- T1490 Inhibit System Recovery (Impact)
- T1110 Brute Force (Credential Access)
- T1078 Valid Accounts (Defense Evasion)
- T1068 Exploitation for Privilege Escalation (Privilege Escalation)
- T1204 User Execution (Execution)
- T1055 Process Injection (Defense Evasion)
- T1059.001 PowerShell (Execution)
- T1027 Obfuscated Files or Information (Defense Evasion)
- T1566 Phishing (Initial Access)

RECOMMENDED ACTIONS

01. Isolate the affected host from the network IMMEDIATELY.
02. Disconnect shared drives and storage resources.
03. Do NOT reboot — preserve memory forensics.
04. Notify incident response team and activate recovery runbook.

The IBM Advantage: Generative BLUF

IBM watsonx.ai



The Technology

IBM watsonx.ai Granite-3-8B-Instruct.

The Application

Generates Bottom Line Up Front (BLUF) natural-language investigation summaries, translating complex correlated data into an instant executive briefing.

Resilience Feature

System features a robust rule-based fallback mode to ensure uninterrupted demonstration continuity if IBM Cloud/API credentials are unavailable.

Technology Stack Blueprint

Frontend

React 18

TypeScript

Vite

Tailwind CSS

Backend

Python 3.11+

FastAPI

Uvicorn

AI & LLM

IBM watsonx.ai
Granite

IBM Bob AI
Assistant

Data & Logic

MITRE ATT&CK
Framework

Simulated SIEM

Deployment & Reproducibility

Zero-Friction Setup

Comprehensive setup guide (docs/setup-guide.md) ensures evaluators can launch both backend and frontend locally in minutes.

Automated Validation

Integrated .github/workflows/validate.yml confirms all structural and metadata requirements are continuously met.

Security Posture

No hardcoded credentials; fully driven by secure .env configurations.

GitHub Actions: Validate Submission Passing

```
> uvicorn main:app --reload  
> npm run dev
```


Roadmap & Honest Limitations

	Date of 80s	Capacity	User	Benefit/loss
1	5023-02-23	secoated	10.0.0.21	53 mls ago
2	2023-03-22	socoated	10.0.0.21	67 mls ago
3	5023-05-23	socoated	10.0.0.21	63 mls ago
4	2023-03-24	socoated	10.0.0.11	87 mls ago
5	5022-03-05	socoated	10.0.0.31	67 mls ago
6	5020-05-06	socoated	10.0.0.31	53 mls ago
7	5023-05-07	secoated	10.0.0.07	54 mls ago
8	5025-03-08	socoated	10.0.0.07	33 mls ago

Current State (Demonstration)

- Simulated alert dataset.
- Heuristic time-window & IP-overlap correlation.
- Local execution without auth/rate limiting.

Future State (Enterprise)

- Live SIEM & EDR data ingestion via API.
- Trained ML clustering models for advanced anomaly correlation.
- Enterprise RBAC (Role-Based Access Control) implementation.

Evaluator Rubric Alignment

✓	Technical Implementation (25pts)	Full-stack, functional application built on React/FastAPI.
✓	Innovation (25pts)	Novel combination of deterministic scoring + AI-assisted BLUF workflow.
✓	Problem Depth (15pts)	Directly solves analyst alert fatigue and manual correlation bottlenecks.
✓	Working Demo (15pts)	Fully reproducible, well-documented local execution + video evidence.
✓	IBM Integration (10pts)	Native integration of IBM Granite and IBM Bob in the core solution pipeline.

Transform Noise Into Knowledge.

End-to-end threat intelligence, accelerated by IBM watsonx.ai.

[View Source Repository]

[Watch Video Demonstration]

[Review Architecture Docs]

Submitted by Team CyberGuard | IBM BoB AI Hackathon 2026